UNIVERSITI TEKNOLOGI MARA

SOFTWARE REQUIREMENTS ANALYSIS FOR SMART ONLINE TUTORING MANAGEMENT SYSTEM (SOTMS)

Version 0.1

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CONFIDENTIAL

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1. Introduction

1.1 Identification

System Number:	SOTMS-2026-04
System Name:	Smart Online Tutoring Management System
System Abbreviation:	SOTMS

1.2 System Overview

This document outlines the purpose and scope of the Smart Online Tutoring Management System (SOTMS). It identifies the main needs and features required to support students, tutors and administrators in managing tutoring activities. The focus is on the essential capabilities that allow users to browse tutors, book sessions and oversee system operations. Detailed explanations of how these requirements are implemented will be provided in the Software Requirements Specification.

1.3 Document Overview

Scope

This Requirements Analysis Document (RAD) applies to the Smart Online Tutoring Management System (SOTMS), which will be developed by the team from Universiti Teknologi Mara (UiTM). The company will develop this web-based system to interface with students, tutors, and administrators.

The SOTMS will be centered around providing a comprehensive platform for managing online tutoring activities, from student registration and tutor browsing to session booking, payment processing, assignment management and schedule coordination. The system will serve universities, colleges and private tutoring agencies for their extra classes.

Purpose

The purpose of this document is to define the high-level requirements of the Smart Online Tutoring Management System (SOTMS) in terms of the needs of the end users. This document outlines the essential capabilities that allow students, tutors, and administrators to effectively manage tutoring activities, including browsing tutors, booking sessions and overseeing system operations

2.0 Referenced Document

2.1 Government Document

Not applicable

2.2 Non-Government Document

[1] SOTMS_Doc1 - Business Process and Activity Diagram

[2] SOTMS_Doc2 - Use Case and System Design

[3] SOTMS_Doc3 - Functional and Non-Functional Requirements Specification

3.0 Functional Requirements

This section defines and describes the functional requirements of the LMS. These requirements are necessary for the user.

Table 3.0: Functional Requirements in SRA

FR No.	Requirement	Description	Phase
SOTMS1_1	Register Account	The system will enable new users (student/tutor) to register an account. The system will securely store user data in the database.	1
SOTMS2_1	Authenticate User	The system shall enable students, tutors and administrators to login to the system by entering valid credentials (username and password)	1
SOTMS2_2		The system will verify entered credentials with the data stored in	1

		the database and grant access to verified accounts.	
SOTMS3_1	Browse Tutors	The system allows student to search and view any available tutors and the system will display the details of the tutor and available tutor slot	2
SOTMS4_1	Book Tutor Session	The system allows students to select a tutor, date and time slot. Booking will be confirmed once payment has been made.	2
SOTMS4_2		The system generates a QR code for the tutor session.	3
SOTMS5_1	Make Payment	The system shall enable students to	1

		make payment for booked tutor session via card, online banking or QR payment	
SOTMS6_1	View Notifications	The system shall generate reminder notifications with session details to students.	2

SOTMS7_1	Participate in Tutoring Session	The system shall display upcoming booked sessions to students	2
SOTMS7_2		The system shall connect students and tutors in a virtual classroom.	2
SOTMS8_1	View Learning Summary and Quiz	The system shall retrieve completed tutoring session data	2
SOTMS8_2		The system shall use AI algorithms to analyze session transcripts, notes or lesson content	3
SOTMS8_3		The system shall automatically generate a personalized learning summary highlighting key concepts covered.	3

SOTMS8_4		The system shall generate interactive quizzes using AI question generation based on the topic discussed.	3
SOTMS9_5	Manage User Accounts	The system shall enable administrators to view and manage data of all users in the system.	1
SOTMS9_6		The system shall enable administrators to approve tutor registrations that are qualified.	2
SOTMS9_7		The system shall enable administrators to remove tutor's accounts who fail to comply with the system's rules and regulations.	2

4.0 Non-Functional Requirements

This section describes the general non-functional requirement for the system. There are as follows:

NFR no	Product Requirement	Description
NFR_1	Performance	● The system shall respond within 1 second for normal operations. ● The system shall handle simultaneous users during peak hours without delay.
NFR_2	Reliability	● The system shall be available 24/7 for booking a session, except for scheduling maintenance. ● The system shall provide data backup and recovery to prevent data loss.
NFR_3	Security	● The system should have authentication to ensure that only authorised users can access the system. ● The system should have an encrypted data mechanism to ensure the confidentiality of student data. ● The system shall enforce strong

		password rules (minimum length, special characters). ● The system shall ensure secure login with session timeout.
NFR_4	Usability	● The system should be user-friendly for easier navigation. ● The system should have guidelines for users to know how to operate the system. ● The system shall provide clear navigation and simple interface for students and tutors. ● The system shall provide error messages and guidance for users.
NFR_5	Maintainability	● The system should be designed to be easily extensible, allowing developers to add new features.
NFR_6	Scalability	● The system must be able to handle a high volume of concurrent users during peak booking hours without crashing.
NFR_7	Availability	● The system shall be highly available to students and tutors to ensure that class

		bookings can occur at any time.
NFR_8	Notification Reliability	• The system shall ensure that notifications are delivered successfully via system and email. • The system shall retry sending notification if delivery fails.

Table 4.0: Non-Functional Requirement in SRA

5.0 Notes

5.1 Introduction

This glossary will be expanded throughout the life of the project.

5.2 Definition

This section describes definitions for all terms used in this document.

5.3 Roles

This chapter describes the roles definition from user point of view.

Student	A user who registers, browse tutors, books tutoring sessions, attends classes.
Tutor	A user who creates and manages their profile, updates availability and conducts tutoring sessions.
Admin	A user who oversees the system, manages student and tutor Accounts, enforces rules and regulations and monitors overall performance.

5.4 Abbreviations and Acronym

This chapter describes the abbreviations and acronyms used in this document and arrange them in alphabetical order.

AI	Artificial Intelligence
FR	Functional Requirements
NFR	Non-Functional Requirements
QR Code	Quick Response Code
SRA	Software Requirement Analysis
UiTM	Universiti Teknologi MARA

6.0 Exhibit

APPENDIX 1

ACTIVITY DIAGRAM

FOR

SMART ONLINE TUTORING MANAGEMENT SYSTEM

ACTIVITY DIAGRAM

SMART ONLINE TUTORING MANAGEMENT SYSTEM

1.1 Description

The diagram describes the process of how students register, book sessions, attend classes, and submit assignments in an online tutoring platform. It illustrates the interactions between four main actors: Student, System, Tutor, and Admin. The workflow includes student registration and session booking, tutor registration and class teaching activities, system automation for scheduling and data management, and administrative functions for approval and payment processing.

1.2 Diagram

Below is the business flow diagram for the online tutoring management system showing the complete workflow across four swimlanes:

- **Student Lane** : Registration, session booking, subject and time selection, class attendance, assignment checking and submission, and payment
- **System Lane** : Automated processes including student ID creation, session scheduling, class link generation, data storage, and submission management
- **Tutor Lane** : Tutor registration, login, scheduling, teaching classes, adding assignments, and grading
- **Admin Lane** : Login, tutor application approval, data access, and payment issuance.

The diagram shows decision points for registration, assignments, approval processes, and data verification, with clear flow paths for successful and failed operations.

ACTIVITY DIAGRAM

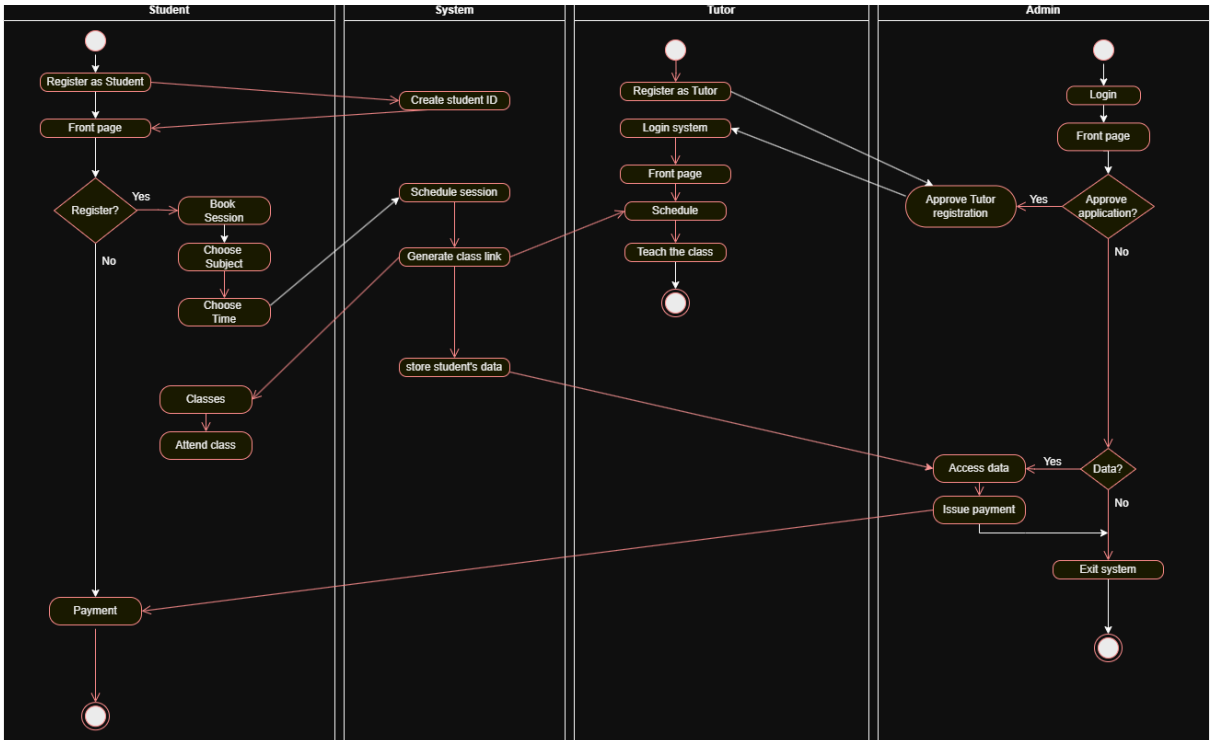


Figure 1.2: Smart Online Tutoring Management System

APPENDIX 2

USE CASE DIAGRAM
FOR
SMART ONLINE TUTORING MANAGEMENT SYSTEM

USE CASE DIAGRAM

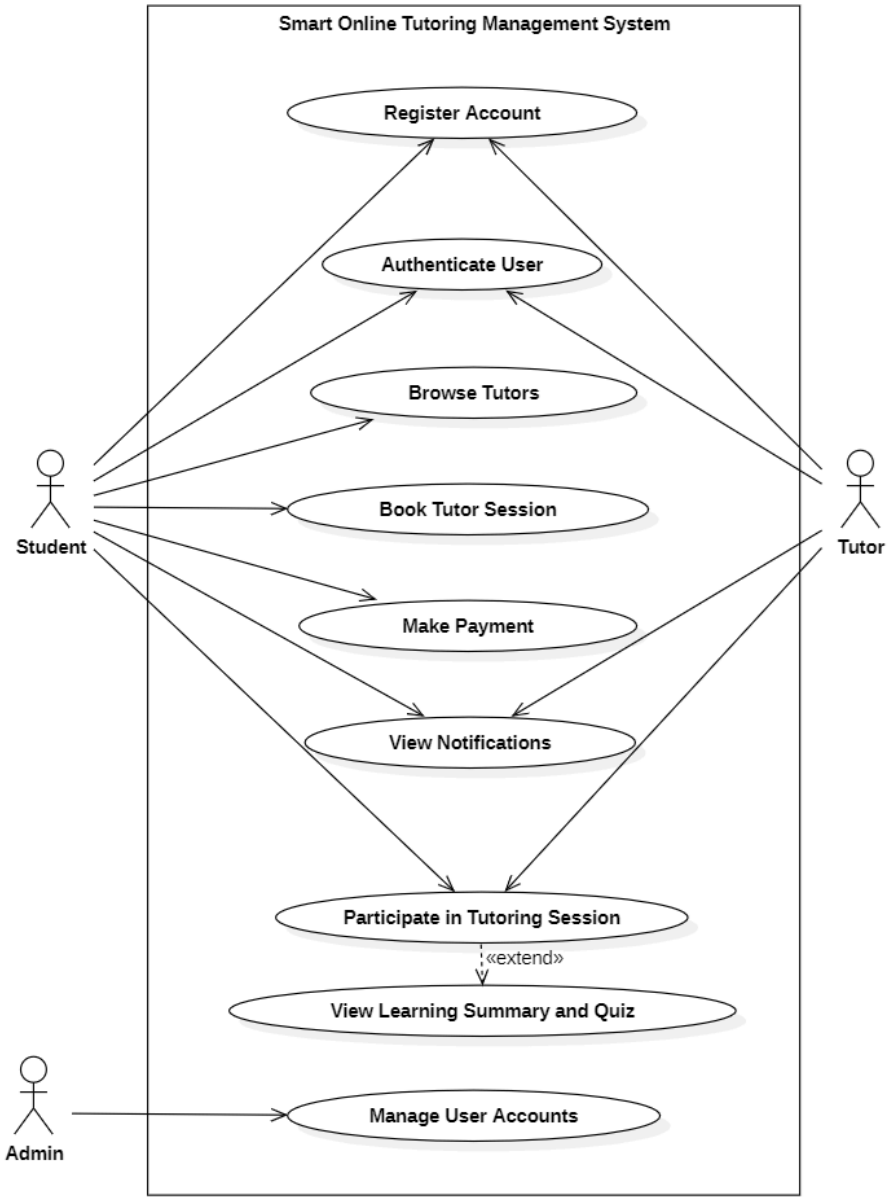


Figure 3.0: SOTMS Use Case Diagram